Note on Superradiant Boson Cloud

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I. SCALAR CLOUD

We use units $G = c = 1$. A complex scalar field Φ with mass μ and minimally coupled to gravity in GR is described by the action [1]

$$S = \int d^4x\sqrt{-g}\left(\frac{R}{16\pi} - \partial^\mu\Phi\,\partial_\mu\bar{\Phi} - \mu^2\Phi\bar{\Phi} + \mathcal{L}_p\right), \quad (1)$$

where $\mathcal{L}_p$ is the Lagrangian density of the point particle representing the secondary in an EMRI, and an overhead bar stands for complex conjugation. The corresponding equations of motion are given by

$$\Box\Phi = \mu^2\Phi\,, \quad (2)$$

$$G_{\mu\nu} = 8\pi\left(T^\Phi_{\mu\nu} + T^p_{\mu\nu}\right), \quad (3)$$

where the stress-energy tensor of the complex scalar field Φ is

$$T^\Phi_{\mu\nu} = 2\partial_{(\mu}\Phi\,\partial_{\nu)}\bar{\Phi} - g_{\mu\nu}\left(\partial_\alpha\Phi\,\partial^\alpha\bar{\Phi} + \mu^2\Phi\bar{\Phi}\right). \quad (4)$$

In this calculation, we choose to solve Eqs. (2) and (3) perturbatively by assuming the characteristic density of the scalar cloud is much smaller than that of the supermassive BH. In this case, we can define

$$\zeta = (\mu M)^3\sqrt{M_c/M}\,, \quad (5)$$

where M_c is the total mass of the cloud and given by

$$M_c = -\int_{r_+}^{\infty}\int_{S^2} g^{t\mu}T^\Phi_{\mu t} \times\left(r^2 + a^2\cos^2\theta\right)dr\,d\Omega\,. \quad (6)$$

We can then perform a two-parameter perturbative expansion of $g_{\mu\nu}$ and Φ as follows:

$$\Phi = \left(\zeta\Phi^{(1,0)} + \mathcal{O}(\zeta^3)\right) + \epsilon\left(\zeta\Phi^{(1,1)} + \mathcal{O}(\zeta^3)\right) + \mathcal{O}(\epsilon^2)\,, \quad (7a)$$

$$g_{\mu\nu} = \left(g^{(0,0)}_{\mu\nu} + \zeta^2 h^{(2,0)}_{\mu\nu} + \mathcal{O}(\zeta^4)\right) + \epsilon\left(h^{(0,1)}_{\mu\nu} + \zeta^2 h^{(2,1)}_{\mu\nu} + \mathcal{O}(\zeta^4)\right) + \mathcal{O}(\epsilon^2)\,. \quad (7b)$$

Under the expansion above, the leading order contribution of $T^\Phi_{\mu\nu}$ occurs at $\mathcal{O}(\zeta^2, \epsilon^0)$, i.e.,

$$G^1_{\mu\nu}\left[h^{(2,0)}_{\mu\nu}\right] = 8\pi\,T^{(2,0)}_{\mu\nu}\,, \quad (8a)$$

$$T^{(2,0)}_{\mu\nu} = 2\partial_{(\mu}\Phi^{(1,0)}\partial_{\nu)}\bar{\Phi}^{(1,0)} - g_{\mu\nu}\left(\partial_\alpha\Phi^{(1,0)}\partial^\alpha\bar{\Phi}^{(1,0)} + \mu^2\Phi^{(1,0)}\bar{\Phi}^{(1,0)}\right), \quad (8b)$$

where $G^1_{\mu\nu}$ is the linearization of the Einstein operator $G_{\mu\nu}$. The definition of ζ in Eq. (5) may seem cyclic, but it means that we evaluate Eq. (8) using $\Phi^{(1,0)}$ with an arbitrary normalization. Since $M_c \propto \Phi^{(1,0)}\bar{\Phi}^{(1,0)}$, this arbitrary normalization is then factored out when we extract the properly normalized $\Phi^{(1,0)}$ using the definition of ζ in Eqs. (5) and (6).

To solve $\Phi^{(1,0)}$, we can first make a harmonic decomposition of $\Phi^{(1,0)}$:

$$\Phi^{(1,0)} = \sum_{\ell_c,m_c} {}_0R^\Phi_{\ell_c m_c\omega_c}(r)\,{}_0S^\Phi_{\ell_c m_c\omega_c}(\theta)\,e^{-\mathrm{i}\omega_c t + \mathrm{i}m_c\phi}\,, \quad (9)$$

where the angular function ${}_0S^\Phi_{\ell_c m_c\omega_c}(\theta)$ is the spin-0 spheroidal harmonics with the spheroidicity

$$\gamma = a\sqrt{\omega_c^2 - \mu^2}\,. \quad (10)$$

One can use the `SpinWeightedSpheroidalHarmonics` package in [2, 3] to compute ${}_0S^\Phi_{\ell_c m_c\omega_c}(\theta)$ with γ in Eq. (10).

From Eq. (2), one can find that [4]

$${}_0R^\Phi_{\ell_c m_c\omega_c}(r\to r_+) = (r - r_+)^{-\mathrm{i}\sigma}\,, \quad (11a)$$

$${}_0R^\Phi_{\ell_c m_c\omega_c}(r\to\infty) = r^{\alpha-1}e^{qr}\,, \quad (11b)$$

where $r_\pm = M \pm \sqrt{M^2 - a^2}$, and

$$\sigma = \frac{2Mr_+(\omega_c - m_c\omega_+)}{r_+ - r_-}\,, \qquad q = -\sqrt{\mu^2 - \omega_c^2}\,, \qquad \alpha = \frac{\mu^2 - 2\omega_c^2}{q}\,, \qquad \omega_+ = \frac{a}{2Mr_+}\,, \quad (12)$$

so one can expand ${}_0R^\Phi_{\ell_c m_c\omega_c}(r)$ as

$${}_0R^\Phi_{\ell_c m_c\omega_c}(r) = (r - r_+)^{-\mathrm{i}\sigma}(r - r_-)^{\mathrm{i}\sigma+\alpha-1}e^{qr} \times\sum_{n=0}^{\infty} a_n\left(\frac{r - r_+}{r - r_-}\right)^n. \quad (13)$$

The coefficients a_n and the quasibound state frequency ω_c can be solved via Leaver's method in [4, 5]. Since we are looking for quasibound states, we require $\mathrm{Im}(\sigma) = 0$. For example, for the fundamental mode of a dipolar cloud ($\ell_c = m_c = 1$, $n_c = 0$) with scalar mass $\mu M = 0.3$ around a Kerr BH of spin $a = 0.877153M$, ω_c is found to be

$$M\omega_c(\ell_c = m_c = 1, n_c = 0) \approx 0.296293\,. \quad (14)$$

This dipolar cloud with the chosen parameters above is also the system we are looking at. In general, given some spin, there will be a corresponding μM that makes $\mathrm{Im}(\sigma) = 0$. This defines a sequence of superradiant boson clouds for Kerr black holes.

One major difference that we have here compared to massive dCS or sGB is that the scalar field is t- and ϕ-dependent; I'm not sure if your code could handle this.

II. METRIC PERTURBATION

A. Coordinates, Tetrad and Gauge

We use ingoing Kerr coordinates $x^\mu = (v, r, \chi, \tilde\phi)$, $v = t + r_*$, $\chi = \cos\theta$, $\tilde\phi = \phi + r_\#$:

$$ds^2 = -\Big(1 - \frac{2Mr}{\Sigma}\Big)dv^2 + 2\,dv\,dr - \frac{4aMr\sin^2\theta}{\Sigma}dv\,d\tilde\phi - 2a\sin^2\theta\,dr\,d\tilde\phi + \frac{\Sigma}{1-\chi^2}d\chi^2 + \sin^2\theta\Big(r^2 + a^2 + \frac{2a^2Mr\sin^2\theta}{\Sigma}\Big)d\tilde\phi^2, \quad (15)$$

where $\Sigma = r^2 + a^2\chi^2$ and $\Delta = r^2 - 2Mr + a^2$. The tetrad is

$$l^\mu = \frac{1}{\Sigma}\Big(r^2 + a^2,\ \frac{\Delta}{2},\ 0,\ a\Big), \qquad n^\mu = (0, -1, 0, 0), \quad (16)$$

$$m^\mu = \frac{1}{\sqrt{2}\,(r + \mathrm{i}a\chi)}\Big(\mathrm{i}a\sin\theta,\ 0,\ -\sin\theta,\ \frac{\mathrm{i}}{\sin\theta}\Big). \quad (17)$$

In this tetrad, we have

$$\rho' = \frac{1}{r - \mathrm{i}a\chi}, \qquad \bar\rho' = \frac{1}{r + \mathrm{i}a\chi}, \quad (18)$$

$$\epsilon' = \frac{\mathrm{i}a\chi}{\Sigma}, \qquad \Psi_2 = -\frac{M}{(r - \mathrm{i}a\chi)^3}. \quad (19)$$

This is the Kinnersley tetrad boosted by $\Lambda = \Delta/(2\Sigma)$,

$$l = \Lambda\, l_K, \qquad n = \Lambda^{-1} n_K, \qquad m = m_K, \quad (20)$$

where, transformed to ingoing coordinates, $l_K = (2(r^2 + a^2)/\Delta, 1, 0, 2a/\Delta)$ and $n_K = (0, -\Delta/(2\Sigma), 0, 0)$. Both l and $n = -\partial_r$ are regular on the future horizon $\mathcal{H}^+$.

We use the traceful outgoing radiation gauge, $h_{ln} = h_{nn} = h_{nm} = h_{n\bar m} = 0$. The remaining components are h_{ll}, $h_{m\bar m}$ (real), $h_{l\bar m}$, $h_{\bar m\bar m}$, and

$$h_{lm} = \overline{h_{l\bar m}}, \qquad h_{mm} = \overline{h_{\bar m\bar m}}. \quad (21)$$

h_{ab} are metric perturbations projected on the tetrads. You may need to adapt the metric into a form that fits your background metric ansatz in METRICS.

Throughout this section h_{ab} stands for $h^{(2,0)}_{ab}$ and T_{ab} for $T^{(2,0)}_{ab}$ of Eq. (8); the linearized Einstein equation is $\delta G_{ab}[h] = 8\pi T_{ab}$.

B. Ψ_4

Ψ_4 obeys $\mathcal{O}_4\Psi_4 = 8\pi\,\mathcal{S}_4 T$ with

$$\mathcal{O}_4 = (\text{þ}' - 4\rho' - \bar\rho')(\text{þ} - \rho) - (\text{ð}' - 4\tau' - \bar\tau)(\text{ð} - \tau) - 3\Psi_2\,. \quad (22)$$

The source built from $\Phi^{(1,0)}\bar\Phi^{(1,0)}$ is independent of v and $\tilde\phi$, so only $\omega = m = 0$ modes are sourced and the spheroidal harmonics reduce to ${}_{-2}Y_{\ell 0}(\chi)$. We write

$$\Psi_4 = W_4 \sum_{\ell\ge 2} R_\ell(r)\,{}_{-2}Y_{\ell 0}(\chi), \quad (23)$$

$$W_4 = \frac{(r + \mathrm{i}a\chi)^2}{(r - \mathrm{i}a\chi)^2\,\Delta^2}. \quad (24)$$

The radial equation is

$$\Delta^2\frac{\mathrm{d}}{\mathrm{d}r}\Big(\frac{R'_\ell}{\Delta}\Big) - \lambda_\ell R_\ell = T_\ell(r), \quad \lambda_\ell = (\ell-1)(\ell+2), \quad (25)$$

$$T_\ell = 2\pi\int_{-1}^{1}\mathcal{T}\,{}_{-2}Y_{\ell 0}\,\mathrm{d}\chi, \qquad \mathcal{T} = -\frac{2\Sigma}{W_4}\,8\pi\,\mathcal{S}_4 T. \quad (26)$$

$R^{\rm in}_\ell$ is the solution for which Ψ_4 is regular on the future horizon, $R^{\rm in}_\ell = O(\Delta^2)$ as $r \to r_+$; $R^{\rm up}_\ell$ is the solution that decays at infinity, $\Psi_4 \propto r^{-\ell-3}$. Dividing Eq. (25) by Δ^2 puts it in Sturm–Liouville form, $(R'/\Delta)' - \lambda_\ell R/\Delta^2 = T_\ell/\Delta^2$, so the Wronskian

$$C_\ell = \frac{R^{\rm in}_\ell\,R^{\rm up\,\prime}_\ell - R^{\rm up}_\ell\,R^{\rm in\,\prime}_\ell}{\Delta} \quad (27)$$

is independent of r.

Variation of parameters gives

$$R_\ell(r) = \frac{1}{C_\ell}\Big[R^{\rm up}_\ell(r)\,I^{\rm in}_\ell(r) + R^{\rm in}_\ell(r)\,I^{\rm up}_\ell(r)\Big], \quad (28)$$

$$I^{\rm in}_\ell = \int_{r_+}^{r}\frac{R^{\rm in}_\ell T_\ell}{\Delta^2}\,\mathrm{d}r', \qquad I^{\rm up}_\ell = \int_{r}^{\infty}\frac{R^{\rm up}_\ell T_\ell}{\Delta^2}\,\mathrm{d}r'. \quad (29)$$

This is the unique solution that is regular on $\mathcal{H}^+$ and decays at infinity.

C. Metric Reconstruction

Let λ be the affine parameter along $n = -\partial_r$ ($\mathrm{d}\lambda = -\mathrm{d}r$). $\int \mathrm{d}\lambda\, X$ denotes the solution Y of $\text{þ}'Y = X$ that vanishes at infinity. Because $\epsilon' \neq 0$, $\text{þ}'Y = -\partial_r Y + (p - q)\,\epsilon' Y$ for Y of GHP type $\{p, q\}$, and

$$Y(r) = \Big(\frac{r + \mathrm{i}a\chi}{r - \mathrm{i}a\chi}\Big)^{\frac{q-p}{2}}\int_r^\infty\Big(\frac{r' + \mathrm{i}a\chi}{r' - \mathrm{i}a\chi}\Big)^{\frac{p-q}{2}}X(r')\,\mathrm{d}r'. \quad (30)$$

This reduces to $\int_r^\infty X\,\mathrm{d}r'$ only for $p = q$. The phase matters for $h_{\bar m\bar m}$ (type $\{-2, 2\}$) and $h_{l\bar m}$ (type $\{0, 2\}$).

The components are obtained in the order below. Each step uses only components already known. $h_{\bar m\bar m}$, $h_{m\bar m}$ and $h_{l\bar m}$ obey second-order equations along n; h_{ll} obeys a first-order one.

1. From Ψ_4:

$$h^p_{\bar m\bar m} = 2\Big[\int \mathrm{d}\lambda\,\frac{\Psi_4}{\bar\rho'} - \bar\rho'\int \mathrm{d}\lambda\,\frac{\Psi_4}{\bar\rho'^{\,2}}\Big]. \tag{31}$$

2. From $\delta G_{nn} = 8\pi T_{nn}$, which contains only $h_{m\bar m}$:

$$h^p_{m\bar m} = \frac{1}{2}\Big[\Big(\frac{\bar\rho'}{\rho'} + \frac{\rho'}{\bar\rho'}\Big)\int \mathrm{d}\lambda\, 4\pi T_{nn}\Big(\frac{1}{\rho'} + \frac{1}{\bar\rho'}\Big) - (\rho' + \bar\rho')\int \mathrm{d}\lambda\, 4\pi T_{nn}\Big(\frac{1}{\rho'^{\,2}} + \frac{1}{\bar\rho'^{\,2}}\Big)\Big]. \tag{32}$$

3. From $\delta G_{n\bar m} = 8\pi T_{n\bar m}$, written as $\mathcal{E}^{l\bar m}_{n\bar m} h_{l\bar m} = \tilde T_{n\bar m}$ with

$$\mathcal{E}^{l\bar m}_{n\bar m} = -\tfrac{1}{2}\big[\text{þ}'(\text{þ}' - 2\rho') + 2\bar\rho'(\rho' - \bar\rho')\big], \tag{33}$$

$$\tilde T_{n\bar m} \equiv 8\pi T_{n\bar m} - \delta G_{n\bar m}\big[h_{\bar m\bar m}, h_{mm}, h_{m\bar m}\big], \tag{34}$$

$$h^p_{l\bar m} = -\frac{1}{3}\Big[\frac{\rho'(\rho' - 2\bar\rho')}{\bar\rho'^{\,3}}\int \mathrm{d}\lambda\,\tilde T_{n\bar m}\,\frac{\rho' + \bar\rho'}{\rho'} - \rho'(\rho' + \bar\rho')\int \mathrm{d}\lambda\,\tilde T_{n\bar m}\,\frac{\rho' - 2\bar\rho'}{\rho'\bar\rho'^{\,3}}\Big]. \tag{35}$$

4. From $\delta G_{ln} = 8\pi T_{ln}$, written as $\mathcal{E}^{ll}_{ln} h_{ll} = \tilde T_{ln}$ with

$$\mathcal{E}^{ll}_{ln} = \frac{r}{\Sigma}\,\partial_r + \frac{r^2 - a^2\chi^2}{\Sigma^2}, \tag{36}$$

$$\tilde T_{ln} \equiv 8\pi T_{ln} - \delta G_{ln}\big[h_{lm}, h_{l\bar m}, h_{mm}, h_{\bar m\bar m}, h_{m\bar m}\big], \tag{37}$$

$$h^p_{ll} = -2(\rho' + \bar\rho')\int \mathrm{d}\lambda\,\frac{\tilde T_{ln}}{(\rho' + \bar\rho')^2}. \tag{38}$$

All integrals run from infinity to r (Eq. (30)) and are evaluated with the pseudospectral code.

Integrating from infinity with $Y \to 0$ sets every homogeneous solution of the transport equations to zero. Since $T_{\mu\nu}$ decays exponentially, no radiative homogeneous piece is needed. What remains are the two perturbations that Ψ_4 does not see: the changes of mass and spin. The full metric perturbation is

$$h_{\mu\nu} = h^p_{\mu\nu} + \delta M\, h^{(M)}_{\mu\nu} + \delta a\, h^{(a)}_{\mu\nu}, \tag{39}$$

where the zero modes are

$$h^{(M)}_{ll} = \frac{2r}{\Sigma}, \tag{40}$$

$$h^{(a)}_{l\bar m} = \frac{\mathrm{i}\sqrt{2}\,Mr\sqrt{1-\chi^2}}{(r - \mathrm{i}a\chi)^3}, \tag{41}$$

$$h^{(a)}_{ll} = -\frac{2Mar(1+\chi^2)}{\Sigma^2}, \tag{42}$$

with h_{lm} the complex conjugate and all other components zero.

The amplitudes are fixed by the cloud's charges,

$$\delta M = M_c, \qquad \delta J = J_c, \qquad \delta a = \frac{\delta J - a\,\delta M}{M}, \tag{43}$$

where, with $\sqrt{-g} = \Sigma$ in $(v, r, \chi, \tilde\phi)$ and the stress tensor T^Φ of Eq. (4),

$$M_c = 2\pi\int_{-1}^{1}\mathrm{d}\chi\int_{r_+}^{\infty}\mathrm{d}r\,\Sigma\,\big(-T^{\Phi\,v}{}_v\big), \tag{44}$$

$$J_c = 2\pi\int_{-1}^{1}\mathrm{d}\chi\int_{r_+}^{\infty}\mathrm{d}r\,\Sigma\,T^{\Phi\,v}{}_{\tilde\phi}. \tag{45}$$

M_c is the cloud mass of Eq. (6): the energy current $-T^{\Phi\,\mu}{}_\nu(\partial_t)^\nu$ is conserved and does not flow into the horizon at $\omega_c = m_c\omega_+$, so the integrals over the slices $t = \text{const}$ and $v = \text{const}$ agree. By fixing the amplitude of the scalar field, we fix $M_c = 10^{-4}M$ and $M = 1$ so that the boson cloud remains a perturbation to Kerr.

In the code, the field is written as $\zeta\Phi^{(1,0)} = A\,\Phi_{\rm unit}$, where $\Phi_{\rm unit}$ has $a_0 = 1$ in Eq. (13) and $\int |{}_0S^\Phi|^2\mathrm{d}\Omega = 1$. Everything ($T_{ab}$, Ψ_4, h^p_{ab}, the zero modes, M_c, J_c) is computed per unit A^2, and $A^2 = M_c/E_{\rm cloud}$ with $E_{\rm cloud}$ given by Eq. (44) for $\Phi_{\rm unit}$. The physical metric perturbation is $A^2 h_{\mu\nu} = \zeta^2 h^{(2,0)}_{\mu\nu}$.

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